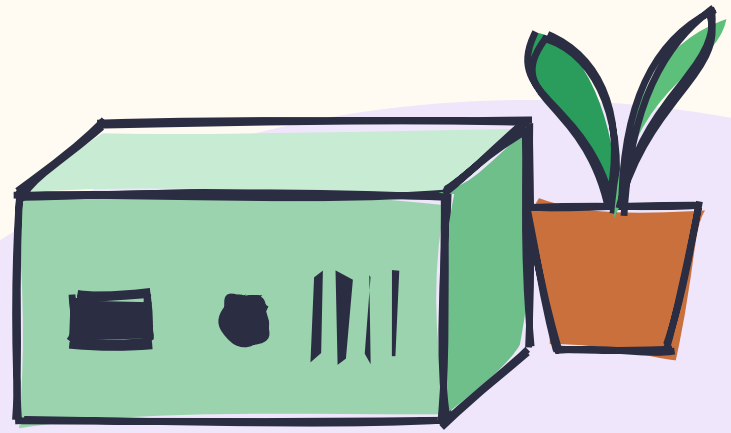
Young Innovators · Class 16

3D-Printed Plant Monitor Enclosure

Turn your circuit into a real product

Design brief | Measuring | Tinkercad | 3D printing



Did you know?
Most electronics you own spent weeks in CAD before anyone printed or moulded a single case.

LEARN

Let's think...

Would you sell a robot made of loose wires?

Bare circuits

Our plant monitor works, but wires fall out, water splashes on the board and it looks like a science experiment. Real products protect their electronics and look good. What would yours need?





Tape it up?

A plastic tub?

Design a case!

Today: you design a **custom enclosure** in Tinkercad, slice it, and 3D print it.

Learning objectives

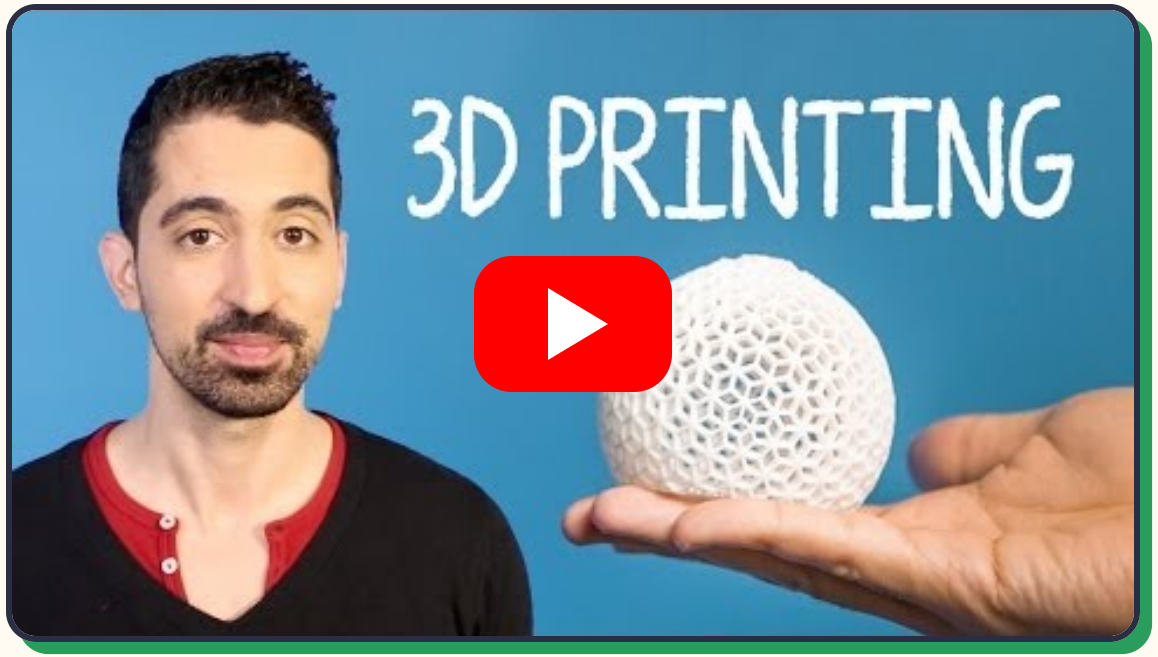
By the end of this class I can...

1	**Design brief:** write **requirements** for a case: what it must protect, fit and let through	**Proof:** my brief lists at least 5 requirements
2	**Measure:** measure parts and add **clearance** so they actually fit	**Proof:** my sketch shows every size with + 0.5 mm clearance
3	**Model:** model a **hollow box, cut-outs and a lid** in Tinkercad	**Proof:** my model has walls, holes and a lid that lines up
4	**Make and test:** **slice** my design, print it and test it against my brief	**Proof:** I ticked or crossed each requirement and wrote one improvement

LEARN

Objective 4 · Make and test · video

Video time!



What Is 3D Printing and How Does It Work?

Mashable Explains · about 2 minutes · youtu.be/Vx0Z6LplaMU

Watch for...

1. How does a 3D printer build an object: **all at once**, or **layer by layer**?
2. What material do most school printers use?
3. Name **two things** people print in the video.

Scan to watch

After the video: Look at your plant monitor. Which part would be easiest to break or get wet? That is your first requirement!

Objective 1 · Design brief

Start with a design brief, not a shape

The idea

A **requirement** is something the case MUST do. Write them before you draw anything:

- Fits the Arduino / ESP32 board and sensor
- USB port reachable for power
- Hole for the soil sensor cable
- Window or hole for the OLED / LED
- Air vents so it doesn't overheat
- Protects from splashes when watering
- Lid opens for repairs

Example: a school bag



A good school bag isn't just 'a bag'. It must fit A4 books, have a bottle pocket, keep rain out, and be strong. Every part of its shape comes from a **need**. Our case is the same.

Real world

Enclosures are designed, not guessed



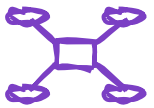
Phone cases
Cut-outs for every button, camera and port



Garden sensors
Sealed against rain and sun (IP65 rating)



Medical devices
Easy to clean, no sharp corners



Drones
Light, strong, with air flow for motors

? Pick up a device in the room. Find 3 design features that solve a problem.

IP rating
IP67 means dust-tight and survives 1 m under water for 30 minutes.

LEARN

Objective 2 · Measure

Measure, then add clearance

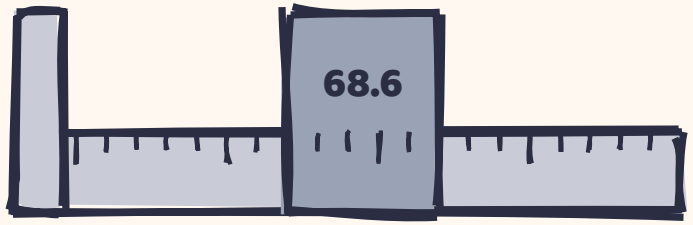
The idea

Measure the board, sensor and OLED with a ruler or **calipers**: length, width, height, and where the USB port and pins are.

Printed parts are never perfect, so add **clearance**: about **+0.5 mm** on each side for parts that slide in. Walls about **2 mm** thick are strong enough.

Example
Arduino Uno: about 68.6 × 53.4 mm. Measure YOUR board anyway: clones differ!

Example: new shoes



Shoes exactly the size of your foot would be impossible to put on. Shops add a little space. Too much, and they fall off. Clearance is the 'just right' gap.

LEARN

Objective 4 · Make and test

Slicing: from 3D model to printer layers

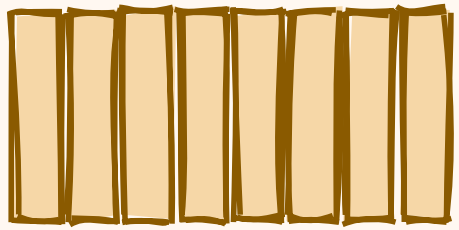
The idea

A **slicer** (UltiMaker Cura) cuts your model into hundreds of thin **layers** (about 0.2 mm) and writes instructions for the printer.

Choose: **PLA**, layer height **0.2 mm**, infill **15–20%**, supports **only if needed**. Read the **print time** and **filament weight** before you print.

The 45° rule
Parts that overhang more than 45° fall down without support. Design flat-bottomed shapes and print them the right way up.

Example: a loaf of bread



0.2 mm layers

A sliced loaf is still the whole loaf, just in thin slices. The printer builds your case one 'slice' at a time, from the bottom up, like stacking bread slices back together.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Reverse engineer
2. Sketch with numbers
3. Model the hollow box
4. Add cut-outs and vents
5. Make a lid that fits
6. Printer checklist
7. Test against your brief

Class game

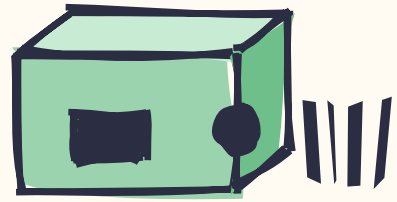
Reverse engineer

How to play

1. Each team gets a real product case: a remote, a charger, a toy, a phone case.
2. In 4 minutes, list every feature: holes, clips, ribs, screws, vents.
3. For each feature, guess the **requirement** behind it.
4. Teams present their best 'aha!' feature.

Then discuss

Which features would you copy for your plant monitor case?



Big idea

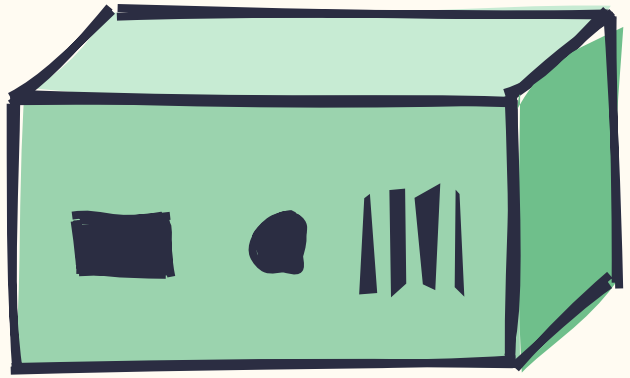
Every hole, clip and rib on a product is there for a reason.

MAKE

Objective 2 · Measure · plan it

Sketch with numbers

Feature	Our size (mm)	Rule
Board (L × W × H)		+1 mm each
Inside of box		board + clearance
Outside of box		inside + 2 × wall
USB hole (W × H, from bottom)		
Sensor cable hole (diameter)		
OLED window (W × H)		



Worked example

Board 68.6 × 53.4 → inside 70 × 55 → outside 74 × 59 with 2 mm walls.

MAKE

Objective 3 · Build step 1 of 3

Model the hollow box

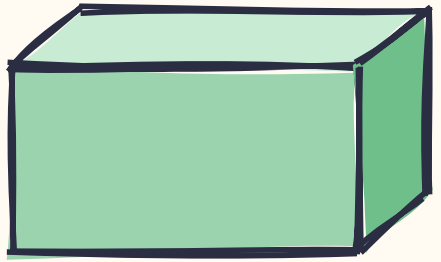
a Go to **tinkercad.com** → 3D Designs → Create. Use your class login.

b Drag a **Box**. Set it to your OUTSIDE size (e.g. 74 × 59 × 35 mm).

c Drag another Box, click **Hole**. Set it to the INSIDE size (70 × 55 × 35).

d Lift the hole **2 mm** up so the floor stays. Centre both with **Align**.

e Select both → **Group** (Ctrl+G). You now have an open box with 2 mm walls.



Check: Looking from above, you see a floor and 4 walls 2 mm thick.

MAKE

Objective 3 · Build step 2 of 3

Add cut-outs and vents

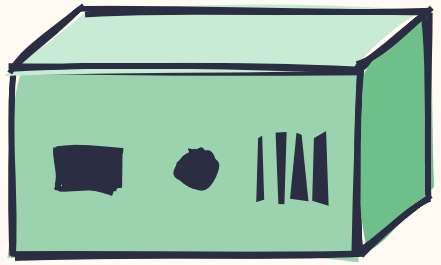
a Use small **Box holes** for the USB port. Position them using your measured height!

b Use a **Cylinder hole** for the sensor cable (e.g. 6 mm).

c Add 4 thin box holes as **vents**, high up on one side so splashes don't get in.

d Add a rectangle hole in the lid area for the OLED window.

e Group again. Spin the view: is every hole all the way through the wall?



Check: Every port and cable has a hole in the right place.

Holes must stick out both sides of the wall, or they will only make a dent.

MAKE

Objective 3 · Build step 3 of 3

Make a lid that fits

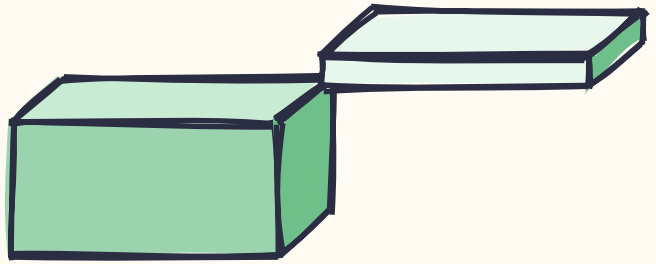
a Copy the outside box. Set its height to **2 mm**: that is the lid.

b Add a **lip** under it: a box 0.5 mm smaller than the inside, 3 mm tall, so it plugs in.

c Emboss your team name or a leaf with the **Text** shape (1 mm tall).

d Move the lid beside the box (not on top) so both print flat.

e Export → **.STL**.



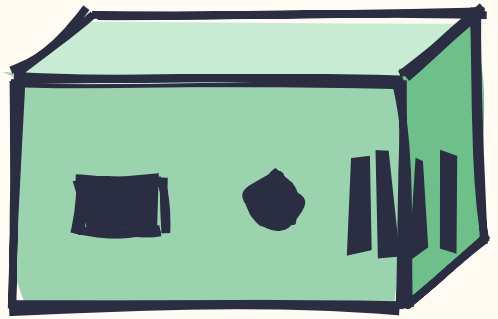
Check: Lid lip slides into the box with a small gap all round.

Lip exactly the inside size? It won't go in. Remember clearance!

Before you press print

Printer checklist

- **Flat on the bed:** the biggest flat face is down.
- **Time:** under 2 hours? If not, make walls thinner or the box smaller.
- **Filament:** note the grams. 1 kg of PLA costs real money: no waste prints.
- **First layer:** watch it. Bad first layer = stop and fix the bed.
- **Safety:** the nozzle is about 200 °C. Hands off while it runs.



Queue

One printer, many teams: print the test-fit piece first (a 5 mm tall slice of the walls).

Test like an engineer

Test against your brief

Requirement	✓ / ✗	Improvement
Board fits and sits flat		
USB cable plugs in		
Sensor cable passes through		
OLED visible through window		
Lid fits snugly		
Survives a splash test (lid on)		

Debug checklist

Too tight? Scale the hole or add 0.3 mm more clearance.
Lid falls off? Make the lip 0.2 mm bigger.

Engineers expect version 2. Every ✗ becomes a change in your next design.

MAKE

Level up!

Designer challenges

Choose your challenge

- **Pot clip:** add a hook so the case clips on the side of a plant pot.
- **Screw posts:** add 4 posts so the board screws down.
- **Drip shield:** a sloped roof so water runs off.
- **Snap-fit:** replace the lip with a clip that clicks shut.

Think like an engineer

Why might a company use injection moulding instead of 3D printing for 10,000 cases?

Boss level

Design a case for next term's robot too, with a battery compartment.

Objective check

Can I really do it? Prove it.

01	Read out 5 requirements from your design brief.	○ ○ ○
02	Show your sketch. What clearance did you add, and why?	○ ○ ○
03	Show your Tinkercad model: the hollow box, the holes and the lid.	○ ○ ○
04	Show your printed case (or test piece). Which requirement failed, and what will you change?	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Product showcase: put the plant monitors in their cases and invite another class to 'buy' the best design.

Show what you know

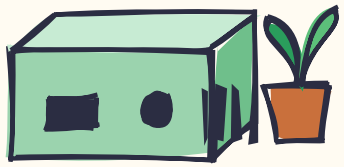
What did we learn?

- [x] Requirements come before shapes
- [x] Measure carefully and add clearance (+0.5 mm)
- [x] A hole object subtracted from a box makes walls
- [x] Slicing turns a model into 0.2 mm layers
- [x] Test against the brief, then make version 2

Exit ticket: your board measures 68.6 mm. How long should the INSIDE of the box be, and why?

At home

Find a case at home (remote, charger, toy). Sketch it and label 3 features with their requirements.



Next class:
Phase 2: AI & coding projects!